Grid Integration of Renewable Energy Sources: Challenges, Issues and Possible Solutions

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***Abstract*—India is considering renewable energy resources (RES) like solar and wind as alternative for future energy needs. As on March 31, 2012 the grid interactive power generation from RES is 24914 MW i.e. around 12.1 % of the total installed energy capacity. Further Ministry of New and Renewable Energy (MNRE), Government of India is targeting to achieve 20000 MW grid interactive power through solar and 38500 MW from wind by 2022. However there are various issues related to grid integration of RES keeping in the view of aforesaid trends it becomes necessary to investigate the possible solutions for these issues. Integration of renewable energy sources to utility grid depends on the scale of power generation. Large scale power generations are connected to transmission systems where as small scale distributed power generation is connected to distribution systems. There are certain challenges in the integration of both types of systems directly. This paper presents the some issues and challenges encountered during grid integration of different renewable energy sources with some possible solutions.**

Keywords—Renewable energy source, grid management, distributed generation

I. INTRODUCTION

Use of Renewable Energy Sources (RES) is increasing drastically due to depletion of fossil fuels and associated environmental problems. Since most of the renewable energy sources are intermittent in nature therefore it becomes a challenging task to integrate RES into the power grid infrastructure [1]. This trends leads to numerous technical and non-technical challenges. For maintaining a reliable and cost effective supply, new efforts have to be undertaken for the management of energy networks, integration of RES and DG in the distribution networks, for generation and load management and for a range of other technical and socio economic aspects of decentralized energy markets. In this paper PV and wind energy sources have been considered under study. The increasing penetration level of PV and wind systems is raising the concerns of some utilities due to the possible negative impacts of the power fluctuations generated from these systems on the network operation [2]. Also, the fluctuation in the power of these systems can lead to unstable operation of the electric network prior to the fault conditions, high power swings in the feeders [3] and unacceptable voltage fluctuations at certain nodes in the electric network [4]. Moreover, the random fluctuations of the power output generated these systems does not allow for considering them in the scheduling process of electricity generation.

This paper is organized as follows: Section II presents the literature review pertaining to the grid integration of renewable energy sources. Challenges and issues are presented in Section III. Some possible solutions are proposed in Section IV. A Conclusion followed by the references is discussed in Section V.

II. LITERATURE REVIEW

In this paper a literature review is carried out related to grid integration of RES. Number of authors/researchers has presented the various issues, challenges and their possible solutions in the context of grid integration. The variability of wind power limits its grid penetration and increases costs. At high penetration levels, the variable output of wind requires utilities to dedicate much of their available reserve generation capacity to accommodate this variability. This paper [5] presents research on improving wind energy integration through more effective coordination of traditional generation resources and energy storage systems that can optimize wind energy production while also increasing the predictability of wind farm output. The increasing use of the Renewable Energy Sources (RES) and the intermittency of the power generated by them create stability, reliability and power quality problems in the main electrical grid [6]. The micro grid is called to be a feasible alternative to solve these issues. As it is a weak electrical grid, the micro grid is very sensitive to load or generation changes. To reduce the effect of these variations and to better harness the energy generated by RES, the Energy Storage Systems (ESS) are used. As the different ESS technologies that are currently available are not enough to satisfy the wide frequency spectrum of the generated energy, the use of a Hybrid Energy Storage System (HESS) is necessary. A HESS is usually formed by two complementary storage devices that can be associated in many different topologies. Of course, the two devices have to be coordinated by an Energy Management System (EMS). In this work the different topologies and energy management algorithms that have been applied in the RES and microgrid contexts have been analyzed and compared. The micro grid can operate both connected to the main grid and in islanding mode. This system is used to overcome the intermittency and uncertainty of the RES. The power converters are used as interfaces between the ESS and the micro grid, to control the power flow of the storage devices and operate the system optimally. However, the power losses of the converters (mainly switching losses) and their economical cost are limiting factors for their use. Although there are methods like the soft-switching designed to reduce the switching losses of the PCS, depending on the application, in some cases it is more economical not to use it.

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A new control strategy for a grid connected doubly fed induction generator (DFIG)-based wind energy conversion system (WECS) is presented [7]. Control strategies for the grid side and rotor side converters placed in the rotor circuit of the DFIG are presented along with the mathematical modeling of the employed configuration of WECS. Battery energy storage system (BESS) to reduce the power fluctuations on the grid due to the varying nature and unpredictability of wind is also presented. Further, the detailed design, sizing, and modelling of the BESS are given for the grid power levelling. Existing control strategies like the maximum power point extraction of the wind turbine, unity power factor operation of the DFIG are also addressed along with the proposed strategy of grid power levelling. An analysis is made in terms of the active power sharing between the DFIG and the grid taking into account the power stored or discharged by the BESS, depending on the available wind energy. This strategy is simulated in MATLAB- SIMULINK and the developed model is used to predict the behaviour.

A case study describing the situation of grid integration of wind energy in German is presented [8]. Due to a new edition of the renewable energy law some technical requirements must be fulfilled by the wind turbines. These demands and the possibilities and efforts of the technical realization are described based on the existing technology. Further, the necessary steps towards an increased power share of wind energy are introduced. This includes the energy storage in power systems and the retrofit of the existing transmission system.

The application of FACTS device (STATCOM) for power quality improvement in grid connected wind generating system and with nonlinear load is presented [9]. The power quality issues and its consequences on the consumer and electric utility are also presented. The operation of the control system developed for the STATCOM in MATLAB/SIMULINK for maintaining the power quality has been simulated. The system has the capability to cancel out the harmonic parts of the load current and maintains the source voltage and current in-phase and support the reactive power demand for the wind generator and load at PCC in the grid system, thus it gives an opportunity to enhance the utilization factor of transmission line. The integrated wind generation and FACTS device with BESS have shown the outstanding performance. The options for larger scale integration of wind power and other renewable energy sources into the grids are demonstrated [10]. In this paper, the German and Greece experience is reported.

The grid frequency deviation caused by wind power fluctuation which is a major concern for secure operation of a power system with integrated large-scale wind power. Many approaches have been proposed to assess this negative effect on grid frequency due to wind power fluctuation [11]. Unfortunately, most published studies are based entirely on deterministic methodology. This paper presents a novel assessment method based on Time-Frequency Transformation to overcome the shortcomings of the existing methods. The main contribution of the paper is to propose a stochastic process simulation model which is a better alternative of the existing dynamic frequency deviation simulation model. In this way, the method takes the stochastic wind power fluctuation into full account so as to give a quantitative risk assessment of grid frequency deviation to grid operators, even without using any dynamic simulation tool. The case studies show that this method can be widely used in different types of wind power system analysis.

Some issues related to grid integration of PV systems have been presented [12]. Further some methods to reduce the fluctuations in PV power output are also discussed. The main issues related to PV systems integration with grid are the fluctuation of PV output power, these fluctuations can negatively impact the performance of the electric networks to which these systems are connected, especially if the penetration levels of these systems are high. Moreover, the fluctuations in the power of PV systems make it difficult to predict their output, and thus, to consider them when scheduling the generating units in the network. In this paper three methods have been investigated such as the use of battery storage systems, the use of dump loads and curtailment of the generated power by operating the power-conditioning unit of the PV system below the maximum power point. The emphasis in the analysis presented in this paper is on investigating the impacts of implementing these methods on the economical benefits that the PV system owner gains. Integration of renewable energy sources (RES) brings to the grid not only their benefits but also some technical problems, as increase of harmonic currents. Compliance with the technical requirements for harmonic current is necessary for successful increase of RES capacity. Scheme to reduce harmonic current for grid-connected PV generation system was developed [13]. This control scheme effectively reduced harmonic current in the grid current of the PV generation system caused by voltage distortions at the grid. Experiments using a prototype of the power conditioning system (PCS) showed its validity. 400 kW PCSs with the control scheme have been installed and have been in service since the end of 2009. In this project, three control methods were developed such as generation power control for fault ride through harmonic current reduction control scheme and grid voltage stabilization using optimal reactive power control. A high performance harmonic current reduction control scheme has been presented. Its purpose is to reduce the harmonic current that flows through a PCS by outputting harmonic voltage equal to the one contained in the grid. The control scheme does not use grid current and has low interference with the existing current control and FRT capability. Extraction algorithm for the harmonic voltages enables the VSI in the PV generation system to output harmonic voltages that are very close to the harmonic voltage of the grid. Experiments showed the validity of the control scheme. 400-kW PCSs with the control scheme were installed at the Hokuto site. The voltage distortions at the Hokuto site were so small that there were not significant differences between the results of the harmonic currents of PCS with and without the control scheme. However, the results measured at the Hokuto site confirm that the harmonic current reduction control does not have negative effects even if there is no large voltage distortion at the grid side.

The distributed generation systems that impose new requirements for the operation and management of the distribution grid, especially when high penetration levels are achieved [14]. In this paper an improved structure of power conditioning system (PCS) for the grid integration of PV solar systems is presented. The topology employed consists of a three-level cascaded Z-source inverter and allows the flexible, efficient and reliable generation of high quality electric power from the PV array. A full detailed model is described and its control scheme is designed. Validation of models and control schemes is carried out through digital simulation using the MATLAB/Simulink environment.

A novel control strategy for achieving maximum benefits from these grid-interfacing inverters when installed in 3-phase 4-wire distribution systems is presented [15]. The inverter is controlled to perform as a multi-function device by incorporating active power filter functionality. The inverter can thus be utilized as:

1) Power converter to inject power generated from RES to the grid, and 2) shunt APF to compensate current unbalance, load current harmonics, load reactive power demand and load neutral current. All of these functions may be accomplished either individually or simultaneously. With such a control, the combination of grid-interfacing inverter and the 3-phase 4-wire linear/non-linear unbalanced load at point of common coupling appears as balanced linear load to the grid. This new control concept is demonstrated with extensive MATLAB/Simulink simulation studies and validated through digital signal processor-based laboratory experimental results. This paper has presented a novel control of an existing grid interfacing inverter to improve the quality of power at PCC for a 3-phase 4- wire DG system. It has been shown that the grid-interfacing inverter can be effectively utilized for power conditioning without affecting its normal operation of real power transfer. This approach thus eliminates the need for additional power conditioning equipment to improve the quality of power at PCC. Extensive MATLAB/Simulink simulation as well as the DSP based experimental results have validated the proposed approach and have shown that the grid-interfacing inverter can be utilized as a multi-function device. Moreover, the load neutral current is prevented from flowing into the grid side by compensating it locally from the fourth leg of inverter. When the power generated from RES is more than the total load power demand, the grid-interfacing inverter with the proposed control approach not only fulfills the total load active and reactive power demand (with harmonic compensation) but also delivers the excess generated sinusoidal active power to the grid at unity power factor. The power quality problems associated with the renewable energy based systems and how custom power devices such as STATCOM, DVR, UPQC play an important role in the power quality improvement are presented [16]. In this paper PV and Wind energy systems integration issues and power quality problems are discussed.

An optimal power management mechanism for grid connected photovoltaic (PV) systems with storage is presented [17]. The objective is to help intensive penetration of PV production into the grid by proposing peak shaving service at the lowest cost. The structure of a power supervisor based on an optimal predictive power scheduling algorithm is proposed. A novel control scheme using a variable frequency transformer (VFT) of 100 MW to effectively reduce power fluctuations of an equivalent 80-MW aggregated doubly-fed induction generator (DFIG)-based offshore wind farm (OWF) connected to an onshore 120-kV utility grid is presented [18]. A frequency-domain approach based on a linearized system model using eigen techniques and a time-domain scheme based on a nonlinear system model subject to disturbance conditions are both performed to examine the effectiveness of the proposed control scheme. It can be concluded from the simulation results that the proposed VFT is effective to smooth the fluctuating active power of the OWF injected into the power grid while the damping of the studied OWF can also be improved. An analysis of the interaction between the variability characteristics of the utility load, wind power generation, solar power generation, and ocean wave power generation is discussed [19]. The results show that a diversified variable renewable energy mix can reduce the utility reserve requirement and help reduce the effects of variability.

The continuously growing amount of renewable sources starts compromising the stability of electrical grids. Contradictory to fossil fuel power plants, energy production of wind and photovoltaic (PV) energy is fluctuating. Although predictions have significantly improved, an outage of multi-MW offshore wind farms poses a challenging problem. One solution could be the integration of storage systems in the grid. After a short overview, this paper focuses on two exemplary battery storage systems, including the required power electronics. The grid integration, as well as the optima usage of volatile energy reserves, is presented for a 5-kW PV system for home application, as well as for a 100-MW medium voltage system, intended for wind farm usage. The efficiency and cost of topologies are investigated as a key parameter for large-scale integration of renewable power at medium and low-voltage. Energy storage systems are a promising solution regarding the integration of fluctuating renewable energy into grids. This paper focuses on small amounts of energy with high peak power, which can be reasonably buffered by battery systems. These energy amounts are required when differences occur between prediction and production of wind or PV power. Two exemplary battery storage systems including the power electronics have been presented. One could state that the modular medium voltage system shows slightly higher converter efficiency, resulting from a multilevel topology. Considering transmission and distribution losses from the medium-voltage grid to the load, the low-voltage solution for domestic use may be just as economically viable because of its relatively high efficiency. However, the relative costs of the low-voltage converter are probably higher due to more overhead. Taking transport losses into account, both systems can achieve nearly equal performance concerning efficiency and commercial viability.

As more renewable power sources, such as wind, solar, and ocean wave, are added to the grid, there has been an increasing impact on the ability of system operators to deal with the added variability that these resources introduce [20]. Since these types of sources are variable and non-dispatchable in nature, their continued integration has required that more and more generating resources be kept in reserve to account for unexpected power changes. While wind, solar, and ocean wave share similarities in their fundamental characteristics as renewable sources, their variability and subsequent impact on the grid can often differ greatly. In combination, however, it has been shown that positive synergistic effects are possible (e.g., in combination, reserve requirements can be reduced compared to single resource scenarios). This paper utilizes the factorial design methodology to analyze the reserve requirement impacts of combining different resources to determine parameters for modeling various grid penetration scenarios. The results point to the possibility of determining an optimal mix of wind, solar, and ocean wave resources. A methodology for determining model to predict the reserve requirements for balancing authority areas with high penetrations of wind, solar, and ocean wave. This model includes interaction effects. The results show that the interaction effects between different renewable sources can result in a lower reserve requirement than would be required if the sources are considered in isolation, although the reduction may be modest. More study on this to find the best mix is necessary. Validation of the presented model with actual data suggests the model is accurate. It is also shown that using a reduced model that ignores interaction effects may be able to provide an acceptable first-order approximation in cases where the interaction effects are not known.

The new power-electronic technology plays a very important role in the integration of renewable energy sources into the grid. It should be possible to develop the power-electronic interface for the highest projected turbine rating, to optimize the energy conversion and transmission and control reactive power, to minimize harmonic distortion, to achieve at a low cost a high efficiency over a wide power range, and to have a high reliability and tolerance to the failure of a subsystem component. The common and future trends for renewable energy systems have been described [21]. As a current energy source, wind energy is the most advanced technology due to its installed power and the recent improvements of the power electronics and control. In addition, the applicable regulations favor the increasing number of wind farms due to the attractive economical reliability. On the other hand, the trend of the PV energy leads to consider that it will be an interesting alternative in the near future when the current problems and disadvantages of this technology (high cost and low efficiency) are solved. Finally, for the energy-storage systems (flywheels, hydrogen, compressed air, super capacitors, superconducting magnetic, and pumped hydroelectric), the future presents several fronts, and actually, they are in the same development level. These systems are nowadays being studied, and only research projects have been developed focusing on the achievement of mature technologies.

New trends in power-electronic technology for the integration of renewable energy sources and energy-storage systems are demonstrated.

The integration of small scale solar and wind generation into distribution networks could cause power fluctuations due to the intermittent and random nature of these sources. This could become one of the significant concern in future power system operator [22]. Though the development of Battery Energy Storage Systems (BESS) can address effectively the aforesaid concern, the high cost of batteries making it economically less attractive. The participation of consumers in energy markets actively and allowing some of their equipment to be used in demand side management operations in future has the potential to reduce the capacity requirements of BESS. Wide scale implementations of demand side management require the exchange of information between controllable system elements and a demand management controller in near real time. A scheme developed for the coordinated control of system elements to reduce fluctuations caused by renewable sources was tested for its performance when information exchange is carried out over a segment of a university network connected to the internet. Results obtained from this study are also presented in the paper. The assessment of the impact of communication over the Internet on the performance of a method to control power systems elements to reduce the power fluctuations was done experimentally. The experimental setup used for simulation studies was created within a laboratory environment.

Some communication technologies available for grid integration of renewable energy resources have been presented in the literature [1]. The communication systems used in a real renewable energy project, Bear Mountain Wind Farm (BMW) in British Columbia, Canada are presented. In addition, the communication systems used in Photovoltaic Power Systems (PPS) is also presented.

III. ISSUES AND CHALLENGES

Renewable energy sources are intermittent in nature hence it is therefore a challenging task to integrate renewable energy resources into the power grid. Some of the challenges and issues associated with the grid integration of various renewable energy sources particularly solar photovoltaic and wind energy conversion systems. Further these challenges are broadly classified into technical and non-technical and described below.

A. Technical Issues

The following are the technical issues are described as

1. Power quality
   a. Harmonics
   b. Frequency and voltage fluctuation
2. Power fluctuation
   a. Small time power fluctuations
   b. Long time or seasonal power fluctuations
3. Storage
4. Protection issues
5. Optimal placement of RES

6. Islanding

Apart from aforesaid technical issues some of the non technical issues are also presented in this paper.

B.Non-Technical Issues

1. Lack of technical skilled man power
2. Less availability of transmission line to accommodate RES
3. RES technologies are excluded from the competition by giving them priority to dispatch which discourage the installation of new power plant for reserve purpose.

IV. POSSIBLE SOLUTIONS

The renewable energy sources such as solar, wind etc. has accelerated the transition towards greener energy sources. The increasing number of renewable energy sources and distributed generators requires new strategies for the operation and management of the electricity grid in order to maintain or even to improve the power-supply reliability and quality. Keeping in view of the aforesaid some of the possible solutions have been proposed by researchers.

1. The power-electronic technology plays an important role in distributed generation and in integration of renewable energy sources into the electrical grid, and it is widely used and rapidly expanding as these applications become more integrated with the grid- based systems. During the last few years, power electronics has undergone a fast evolution, which is mainly due to two factors. The first one is the development of fast semiconductor switches that are capable of switching quickly and handling high powers. The second factor is the introduction of real- time computer controllers that can implement advanced and complex control algorithms. These factors together have led to the development of cost- effective and grid-friendly converters.
2. Intermittence of power generation from the RES can be controlled by generating the power from distributing the RES to larger geographical area in small units instead of large unit concentrating in one area. For example output power of large solar PV system with rating of tens of mega watt can be change by 70% in a five to ten minutes of time frame by the local phenomenon like cloud passing etc therefore large number of small solar PV system should be installed in larger geographical area. The fluctuation of total output power can be minimized because of local problem can affect only small unit power not the total output power.
3. In case of irrigation load the load is fed during the night time or off peak load time and this is fed by conventional grid. On other hand power generated by RES like solar PV is generated during day time so we can use this power for irrigation purposes instead of storing the energy for later time which increases the cost of the overall system. Using the solar water pumping for irrigation gives very high efficiency approx 80% to 90% and the cost of solar water pumping is much lesser than the induction motor pumping type.
4. In large solar PV plant output power is fluctuating during the whole day and this power is fed to the grid and continuously fluctuating power gives rise to the security concern to the grid for making stable grid. Solar PV plant owner have to install the different type of storage system which gives additional cost to the plant owner. Once the storage system is fully charged then this storage elements gives no profit to the system owner. Therefore solar based water pumping system may be installed instead of storage system.

V. CONCLUSIONS

In this paper, some issues related to grid integration of RES and their possible solutions available in the literature have been presented. To minimize the fluctuations and intermittent problems power electronics devices are the viable options. Further, energy storage and use of dump load and MPPT could be used for reducing the power fluctuations in PV systems. In addition to the aforesaid, the up gradation in balance of systems by incorporating the new materials and storage elements could reduce the problems associated with grid integration.

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